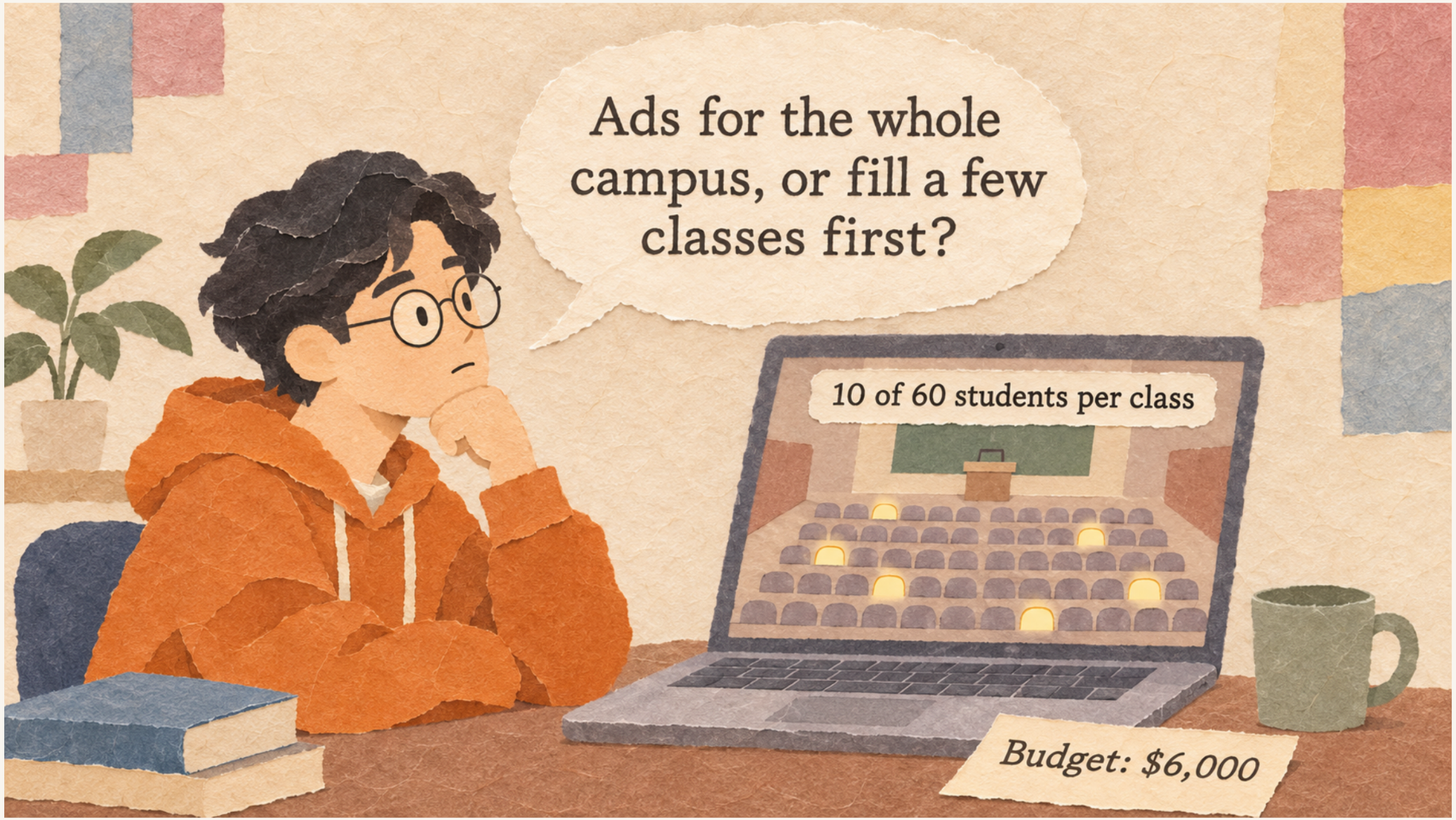
Session 1.2: The Network Effect Strategy



THE SITUATION

ClassChat: 3,000 users, but every class feels empty

ClassChat is Tomás's free app where students in the same class chat, share notes, and ask questions before exams. Users are spread over 300 classes, so most students open an almost empty class and leave.

CLASSCHAT (THIS SEMESTER)	VALUE
Users / classes with at least one user	**3,000 / 300**
Students in a typical class / users in it	**60 / 10**
Weekly active rate, classes with 30 or more users	**80%**
Weekly active rate, classes with fewer than 30 users	**15%**
Money to spend on growth	**$6,000**

What is the Network Effect Strategy?

Some products become **more useful for each user when more people use them**: a chat app is useless if none of your friends use it. Nobody wants to join first, and an almost empty product loses its first users. So you **fill one small group first** until it is clearly useful there, give early users a reason to stay, help people join together, and check with data that each new user really makes the product better. Then you move to the next group.

Use it when users get value from each other, not only from the product itself.

Network effect, critical mass, penguin problem

Network effect

A product becomes more useful for each user as more people use it.

Example: a chat app is more useful when more classmates use it.

Direct network effect

Users gain from users of the same group. (Session 1.3: two different groups.)

Example: chat app users want other chat app users.

Critical mass

The number of users at which a product is useful enough that people stay and invite others.

Example: about 30 of the 60 students in a class.

Penguin problem and density

Penguin problem

Everyone waits for someone else to join first, so nobody joins.

Example: penguins on the ice wait for one to jump into the water first.

Density

How full a group is. Density = users in the group / people in the group x 100.

Example: 12 users in a class of 60: 20%.

Standalone value and weekly active rate

Standalone value

The value a user gets even when no one else uses the product.

Example: a note app that organizes your own notes.

Weekly active rate

Weekly active users / all users x 100.

Example: 45 of 300 users: 15%.

Rule 1: Fill one small group first

A product feels alive when a group is full, not when many groups each have a few users.



DO

Example

ClassChat spends its budget on 40 first-year business classes, so most students there join at once.

AVOID

Common mistake

Advertising to the whole campus. Users are spread thin, no class feels alive, and new users leave.

Rule 2: Give the first user value even when alone

Early users join an almost empty product. Give them a reason to stay until their friends arrive.



DO

Example

ClassChat organizes each student's own timetable and notes, so it is useful from the first day.

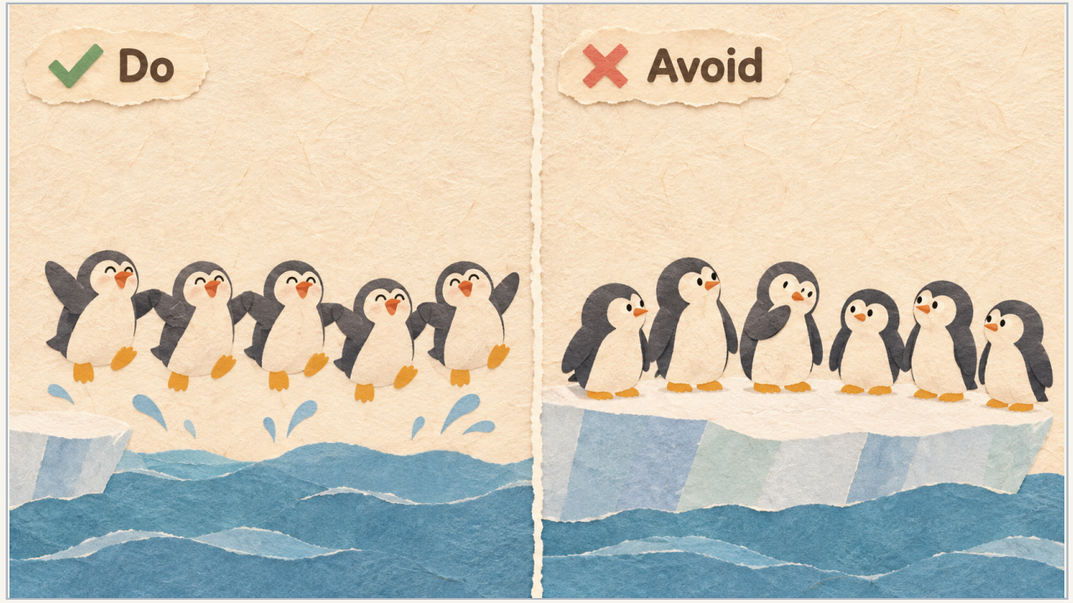
AVOID

Common mistake

A product that is useless until friends arrive. The first users delete it before friends come.

Rule 3: Help the first penguins jump together

People join when they see others join. Bring a whole group in at the same moment.



DO

Example

A class ambassador asks everyone to join in the first lecture, so 40 students join on the same day.

AVOID

Common mistake

Letting people join one by one, or limiting invitations so friends cannot get in.

Rule 4: Check that new users make it better for others

Do not count downloads only. Measure whether fuller groups are more active.



DO

Example

80% weekly active where 30 or more students joined; 15% where fewer joined. The effect is real.

AVOID

Common mistake

Celebrating downloads. Many people install the app once and never come back.

Facebook

What happened

Facebook started on February 4, 2004, open only to Harvard students. A month later it opened to just three more universities (Columbia, Stanford, and Yale), then to other universities one by one, before opening to everyone in 2006.

Why it is good

- It filled one university at a time, so each campus was full of real friends. **(Rule 1)**
- Students at a campus joined together, because their friends had just joined. **(Rule 3)**
- It opened a new campus only when earlier ones were active. **(Rule 4)**

Sony PlayStation 3 and Blu-ray

What happened

Around 2006, two movie-disc formats competed: Blu-ray (Sony) and HD DVD (Toshiba). Buyers and film studios waited to see which would win. Sony put a Blu-ray player inside every PS3. About 10.5 million PS3s were sold by early 2008, Warner Bros. chose Blu-ray only, and in February 2008 Toshiba stopped making HD DVD.

Why it is good

- Every PS3 buyer became a Blu-ray owner at once: millions jumped together. **(Rule 3)**
- The PS3 was useful as a game console even without movies. **(Rule 2)**
- More owners made studios choose Blu-ray, which helped every owner.

Google Wave

What happened

Google Wave was a tool for writing and chatting together online. In September 2009, Google opened it to about 100,000 invited users. Most could not invite their friends or coworkers, and many did not understand what to use it for. In August 2010, Google stopped developing it.

Why it is bad

- Limited invitations: users could not bring the people they worked with. **(Rule 3)**
- A tool for working together is useless alone. **(Rule 2)**
- Users were spread over the world instead of whole teams joining. **(Rule 1)**

BAD EXAMPLE 2

Clubhouse

What happened

Clubhouse is an app for live audio conversations. It grew very fast in early 2021 and had about 14 million downloads by mid-April 2021. But many users found few good live rooms, other apps added live audio, and use fell sharply. In April 2023, Clubhouse cut more than half of its staff.

Why it is bad

- Growth was measured by downloads, not by users who came back. **(Rule 4)**
- With no interesting room live, the app gave a user nothing to do. **(Rule 2)**
- Users could easily listen on other apps too (Session 1.4).

Which rule made the difference?

EXAMPLE	RESULT	THE RULE THAT DECIDED IT
Facebook	**Good**	One campus at a time (Rule 1)
PS3 and Blu-ray	**Good**	Millions joined together (Rule 3)
Google Wave	**Bad**	Friends could not get in (Rule 3)
Clubhouse	**Bad**	Counted downloads, not return visits (Rule 4)

A full small group beats a large empty network.

Tool 1: Density

USED IN EXERCISES 2 AND 3

Formula

Density = users in the group / people in the group x 100

WHAT IT LOOKS LIKE

Example: a dorm

- 200 students; 50 use a laundry-sharing app.
- 50 / 200 x 100 = 25%.

Tool 2: Weekly active rate

USED IN EXERCISES 2 AND 3

Formula

Weekly active rate = weekly active users / all users x 100

WHAT IT LOOKS LIKE

Example

- 50 users; 35 use it every week.
- 35 / 50 x 100 = 70%.

Tool 3: Cost per new user

USED IN EXERCISE 3

Formula

Cost per new user = money spent / new users

WHAT IT LOOKS LIKE

Example: a game

- $2,000 on ads brings 500 new players.
- $2,000 / 500 = $4.00.

Tool 4: Trade-off and decision rule (from Session 1.1)

MEMO BULLET 2

Trade-off

What you give up to get something you value more.

MEMO BULLET 3

Decision rule

A promise, written before you start, about which result would make you change your decision.

Example: "If fewer than 30 students in a class join by week 3, we will stop spending in that class."

Is there a network effect?

Six products:

- (a) A messaging app.
- (b) A calculator app.
- (c) An online game where players play against each other.
- (d) A flashlight app.
- (e) A language-exchange app that matches learners who teach each other.
- (f) A paid photo-editing app.

Use the template

TEMPLATE

Product [letter]: [network effect / no network effect], because when one more user joins,
[who gains what / nobody else gains anything].

WORKED EXAMPLE: TWO OTHER PRODUCTS

A video-call app: network effect, because when one more user joins,
everyone who knows that person can now call them.
An alarm clock app: no network effect, because when one more user joins,
nobody else gains anything.

Does one more user help the others?

PRODUCT	ANSWER	WHY
(a) Messaging app	Network effect	More people you can message
(b) Calculator	No	Works the same alone
(c) Online game	Network effect	More players, faster matches
(d) Flashlight	No	Works the same alone
(e) Language exchange	Network effect	More partners for each language
(f) Photo editor	No	Works the same for each user

Find critical mass: GymBuddy

GYM	MEMBERS	GYMBUDDY USERS	WEEKLY ACTIVE
A	**400**	**40**	**6**
B	**500**	**200**	**150**
C	**300**	**150**	**105**
D	**600**	**60**	**9**
E	**200**	**30**	**6**

- **Step 1:** For each gym, density and weekly active rate (Tools 1 and 2).
- **Step 2:** At what density does the weekly active rate jump?
- **Step 3:** How many more users does each low gym need? Which gym is cheapest to fill?

EXERCISE 2 / MEDIUM / 30 MIN

Use the template

TEMPLATE

Critical mass is about [number]% density, because groups above it
have [number]% weekly active users and groups below it only [number]%.
We should focus on [group] next, because it needs only [number] more users.
We will use Rule [number]: [one action].

WORKED EXAMPLE: STUDYBUDDY IN FOUR DORMS

Critical mass is about 50% density, because dorms above it have
75-80% weekly active users and dorms below it only 20%.
We should focus on Dorm 3 next, because it needs only 40 more users
(Dorm 2 needs 160).
We will use Rule 3: invite a whole floor to a study night.

Critical mass is about 40%; fill Gym E next

GYM	DENSITY	WEEKLY ACTIVE RATE	USERS NEEDED FOR 40%
A	10%	15%	120
B	40%	75%	-
C	50%	70%	-
D	10%	15%	180
E	15%	20%	50

A strong answer names one action, such as a "find a partner" week where a whole class joins together (Rule 3).

Where should ClassChat spend $6,000?

300 classes of 60 / about 10 users each / 80% active at 30+ users, 15% below / both options: $6,000, 1,500 new users

SPREAD WIDE

Option A: Ads for the whole campus

The 1,500 new users spread evenly over all **300** classes.

FILL ONE GROUP

Option B: Fill 40 classes first

$100 ambassador in each of **40** first-year business classes ($4,000) and a launch event ($2,000).

All 1,500 join in those 40 classes, which now have **400** users.

Calculate before you decide (Tools 1, 2, and 3)

- Cost per new user in both options = $6,000 / 1,500 = $_____
- Now: weekly active users = 3,000 x 15% = _____
- Option A: users per class = 4,500 / 300 = _____; density = _____%; weekly active = 4,500 x _____% = _____
- Option B: users per class in 40 classes = (400 + 1,500) / 40 = _____; weekly active there = 1,900 x _____% = _____; other 260 classes = 2,600 x 15% = _____; total = _____
- New users Option B needs so that the 40 classes reach 30 each: 40 x 30 - 400 = _____

Write a 3-bullet memo to Tomás (Tool 4)

TEMPLATE

To: [founder name], [company]
1. Decision and evidence: We should choose [option] because [the rule it follows] and [one number from your calculation].
2. Biggest risk and trade-off: The weakest point is [risk]. We give up [what we lose] to gain [what we protect or win].
3. Next step and decision rule: In the first weeks we will [one action] and measure [one number]. We will change this decision if [result].

Then give a 2-minute presentation using your three bullets.

GymBuddy: focus on Gym E?

To: Sam Lee, founder, GymBuddy
1. Decision and evidence: We should focus next month's budget on Gym E, because gyms at 40% density have 70-75% weekly active users (Rule 4) and Gym E needs only 50 more users to get there (Rule 1).
2. Biggest risk and trade-off: The weakest point is that Gym E is small, so it adds few users. We give up faster growth in user numbers to make one more gym truly active.
3. Next step and decision rule: In the next four weeks we will run a "find a partner" week at Gym E and measure density. We will change this decision if Gym E is still below 30% density after four weeks.

Same cost, almost 3 times more active users

- Cost per new user = **$4.00** in both options. Now: **450** weekly active users.
- Option A: **15** users per class, density **25%** (below critical mass); 4,500 x 15% = **675**.
- Option B: **47.5** users per class; 1,900 x 80% = 1,520, plus 2,600 x 15% = 390; total = **1,910**.
- Option B needs at least **800** new users in the 40 classes. With only 600, each class has 25 users, and the total falls to 3,600 x 15% = 540.

What a strong memo looks like

- Chooses Option B (Rules 1 and 3): the cost is the same, so the difference is where the users go.
- Names the risk: ambassadors may bring fewer than 800 users.
- Decision rule: "If fewer than 20 students in a class join by week 3, we will move that ambassador budget to another class."

SUCCESS CHECK

Can you do these three things?

01

Explain network effects, critical mass, and the penguin problem.

02

Measure density and weekly activity to find critical mass.

03

Compare "spread wide" with "fill one group first" and write a decision rule.

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